

Mount Kenya University

Unlocking Infinite Possibilities

SCHOOL OF COMPUTING AND INFORMATICS

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

UNITE TITLE: ADVANCED WEB DESIGN AND DEVELOPMENT

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PROJECT TITLE: MOONLIGHT ENTERPRISES

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1.0 INSTALLATION LOCALHOST TOOL

1.1 XAMPP

XAMPP was installed successfully to create a local server environment for web development. Apache and MySQL services were included to help run websites and manage databases on the computer.



Figure 1:XAMPP INSTALLATION

1.2 Tested localhost successfully

The localhost was tested by opening `http://localhost` in the browser. The XAMPP dashboard appeared successfully, showing that the Apache server was running correctly and the local environment was ready for web development.



Figure 2:LOCALHOST TESTING

1.3 Created Hello World pages



Figure 3:CODE



Figure 4: CREATION OF HELLO WORLD

1.4 Tested basic database connectivity

A database connection test was performed using PHP and MySQL. The successful connection message confirmed that the web application could communicate properly with the database server.

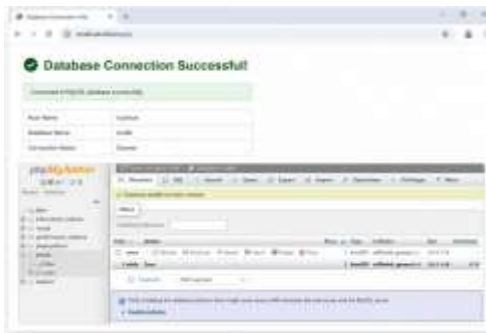


Figure 5: TESTING DATA CONNECTION

REFLECTION

I successfully installed XAMPP and configured the localhost environment for web development. This helped me understand how local servers work and how to run web applications on my computer.

2.0 Website Design and Project Planning

2.1 Designed Wireframes

Basic sketches were created to show the structure and layout of the website before development.



Figure 6: DESIGNED FRAME

2.2 Designed GUI Layouts

User interface designs were created to show how the web pages would look and function.



Figure 7:Designed GUI Layouts

2.3 Planned Project Workflow

The steps and activities of the project were organized to ensure smooth development.

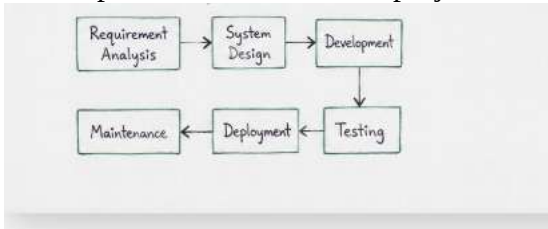


Figure 8:Planned Project Workflow

2.4 Selected Technologies

Appropriate tools and technologies such as HTML, CSS, PHP, MySQL, and XAMPP were chosen for the project development.

REFLECTION

I designed wireframes, GUI layouts, and planned the workflow of the project before development. This improved my skills in organizing and planning a complete web application project.

3.0 MOONLIGHT POS SYSTEM REPORT

Moonlight POS is a web-based point of sale and Inventory Management System developed using PHP, MySQL, HTML, CSS and JavaScript. The system manages inventory, sales records and transaction tracking.

3.1 Login Page



Figure 9:LOGIN PAGE

3.2 Dashboard



Figure 10:DASHBOARD

REFLECTION

I developed the Moonlight POS system with a login page and dashboard using PHP, HTML, CSS, JavaScript, and MySQL. This helped me gain practical experience in creating a functional web-based system.

4.0 INVENTORY MANAGEMENT(CRUD)

Inventory Module

The inventory section stores and manages product information such as item name, stock code, quantity, supplier and pricing details.

4.1 Stock Inventory Page



Moonlight POS

Current Stock Inventory

Default view of all items, quantities, and pricing. Items with quantity less than or equal to 0 are flagged.

ID	Item Name	Stock Code	Quantity	Purchase Price	Selling Price	Supplier	Date Received	Received By	Status
21	Sabud	QABR00	300	Ksh 200.00	Ksh 280.00	Aburadish	2025-10-11	Ido	✓ Edit ✕ Delete
18	Moro SD 4000	QWGR01	145	Ksh 4.00	Ksh 13.00	WonggPro	2025-10-17	Ido	✓ Edit ✕ Delete
15	HERB Soda 200	HEBES20	75	Ksh 5.00	Ksh 12.00	Chidafay	2025-10-17	Ido	✓ Edit ✕ Delete
22	Ward	W04	65	Ksh 300.00	Ksh 500.00	Ido	2025-10-11	@mama	✓ Edit ✕ Delete
27	USB Cable	USB0040	45	Ksh 10.00	Ksh 20.00	WonggPro	2025-10-18	Ido	✓ Edit ✕ Delete
14	Pepsi Soda 180	PEPS018	40	Ksh 25.00	Ksh 40.00	Motokidsh	2025-10-17	Ido	✓ Edit ✕ Delete
19	Soda 170	SD0001	85	Ksh 4,000.00	Ksh 3,000.00	Ido (not)	2025-10-18	Ido	✓ Edit ✕ Delete

Figure 11: Stock Inventory Page

4.2 CRUD OPERATIONS

The system implements CRUD operations:

1. Create -add new products.
2. Read -view inventory records.
3. Update -edit product information.
4. Delete -remove unwanted records.

4.3 Add Item



Moonlight POS

+ Add New Item to Inventory

Use this form to add a brand new product to your stock list.

Item Name:

Unique Stock Code (SKU):

Initial Stock Quantity:

Purchase Price (Cost):

Selling Price (Retail):

Supplier:

Figure 12: ADD ITEM

4.4 Delete function



Moonlight POS

Current Stock Inventory

Default view of all items, quantities, and pricing. Items with quantity less than or equal to 0 are flagged.

✕ Error: Cannot delete item ID 21. It has 4 associated sales records. Please delete the associated sales records first.

ID	Item Name	Stock Code	Quantity	Purchase Price	Selling Price	Supplier	Date Received	Received By	Status
21	Sabud	QABR00	300	Ksh 200.00	Ksh 280.00	Aburadish	2025-10-11	Ido	✓ Edit ✕ Delete
18	Moro SD 4000	QWGR01	145	Ksh 4.00	Ksh 13.00	WonggPro	2025-10-17	Ido	✓ Edit ✕ Delete
15	HERB Soda 200	HEBES20	75	Ksh 5.00	Ksh 12.00	Chidafay	2025-10-17	Ido	✓ Edit ✕ Delete
22	Ward	W04	65	Ksh 300.00	Ksh 500.00	Ido	2025-10-11	@mama	✓ Edit ✕ Delete
27	USB Cable	USB0040	45	Ksh 10.00	Ksh 20.00	WonggPro	2025-10-18	Ido	✓ Edit ✕ Delete

Figure 13: DELETE FUNCTION

4.5 Update



Figure 14:UPDATE

REFLECTION

I implemented CRUD operations to add, view, update, and delete inventory records. Through this, I learned how to manage database records dynamically in a web application.

5.0 SALES MANAGEMENT

5.1 Record New Sale

The sales module allows users to record complete sales transactions into the system database.



Figure 15:Record New Sale

5.2 Sales Records

The system displays transaction history including quantity sold, sale amount, date and salesperson information.

Item ID	Item Name	Stock Code	Qty Sold	Unit Price (per Item)	Total Sale	Date & Time	Sold By
10	ITEM-476	ITEM476	4	Ksh 100,000.00	Ksh 400,000.00	2025-10-20 12:34	John
11	Laprop	LAP123	1	Ksh 20,000.00	Ksh 20,000.00	2025-11-01 16:39	John
14	Laprop	LAP123	24	Ksh 946,750.00	Ksh 22,722,000.00	2025-11-01 11:39	John
17	ITEM4 Cofee 2m	ITEM4C2M	6	Ksh 12.00	Ksh 72.00	2025-11-01 11:19	John
12	More 50-6403	506403	1	Ksh 200.00	Ksh 200.00	2025-10-21 11:04	John
11	Tobacco	Tob100	1	Ksh 200.00	Ksh 200.00	2025-10-01 12:17	John
10	Tobacco	Tob100	7	Ksh 200.00	Ksh 1,400.00	2025-10-01 12:14	John
8	External HDD	HDD750	1	Ksh 95.00	Ksh 95.00	2025-10-01 11:49	John
8	Keyboard	KEY789	1	Ksh 25.00	Ksh 25.00	2025-10-01 11:39	John
7	Internal HDD	HDD750	6	Ksh 83.00	Ksh 500.00	2025-10-01 11:13	John
8	Keyboard	KEY789	1	Ksh 15,000.00	Ksh 15,000.00	2025-10-14-09:28	John
7	Tobacco	Tob100	6	Ksh 78.00	Ksh 468.00	2025-10-11 18:22	John
8	Tobacco	Tob100	7	Ksh 36.00	Ksh 252.00	2025-10-12 18:22	John

Figure 16: Sales Records

REFLECTION

I created features for recording sales and viewing sales records. This improved my understanding of transaction management and data handling within the system.

6.0 DATABASE AND GITHUB

6.1 Database Integration

The system uses PHP and MySQL database connection to store, retrieve, update and manage data dynamically.

6.2 phpMyAdmin Database Table

Item ID	Item Name	Stock Code	Qty Sold	Unit Price	Total Sale	Date & Time	Sold By

Figure 17: phpMyAdmin Database Table

6.3 Database connection code

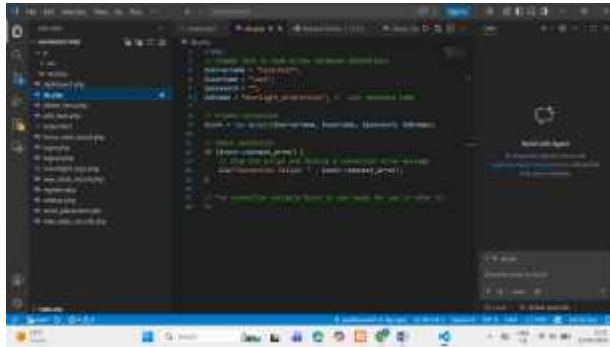


Figure 18: Database connection code

6.4 GitHub Version Control

GitHub was used to save project progress, maintain backups and track development updates.

6.5 GitHub Repository System

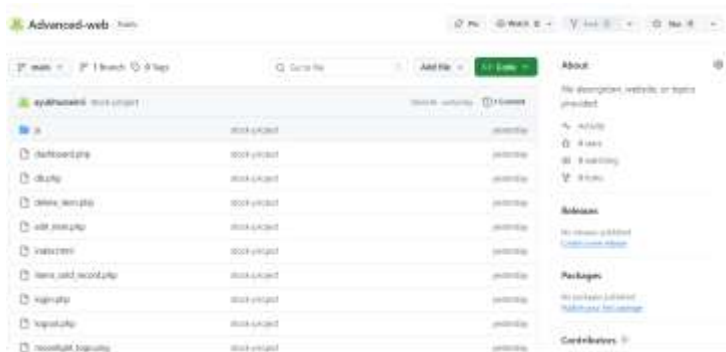


Figure 19:GitHub Repository System

<https://github.com/ayubhussein5/Advanced-web>

REFLECTION

I connected the system to a MySQL database and used GitHub for version control and backups. This helped me understand database integration and the importance of managing project versions online.

7.0 CONCLUSION

Moonlight POS successfully demonstrates backend development, database integration, authentication, inventory management, sales processing and CRUD operations.

8.0 REFERENCE

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